

Problem Statement: Student Record Management Using Tuples

A coaching institute wants to store student information in a way that prevents accidental modification of records.

Each student's record should contain:

- Student ID
- Student Name
- Course Name
- Fees Paid

Store the details of 5 students using tuples and perform the following operations:

1. Display all student records.
2. Display the first student's details.
3. Display the last student's details using negative indexing.
4. Display only Student ID and Name for all students.
5. Count the total number of students.
6. Check whether a student named "Rahul" exists in the records.

Sample Data

```
students = (  
    (101, "Rahul", "Python", 25000),  
    (102, "Priya", "Java", 30000),  
    (103, "Amit", "Data Science", 45000),  
    (104, "Sneha", "Python", 25000),  
    (105, "Rohan", "MERN", 40000)  
)
```

Classwork – Tuples in Python

Question 1: Employee Performance Evaluation

Problem Statement

A company stores employee details in a tuple. Each employee record contains:

```
employees = (  
    ("E101", "Anuj", 92),  
    ("E102", "Rahul", 76),  
    ("E103", "Priya", 58),  
    ("E104", "Neha", 88),  
    ("E105", "Amit", 45)  
)
```

Where:

- First value = Employee ID
- Second value = Employee Name
- Third value = Performance Score

Tasks

Write a Python program to:

1. Display details of employees scoring **80 or above**.
2. Count the number of employees who need improvement (score below 60).
3. Find the employee with the highest score.
4. Create a list containing the names of all employees scoring above 75.
5. Display the performance category for each employee:
 - 90 and above → Excellent
 - 75 to 89 → Good
 - 60 to 74 → Average
 - Below 60 → Needs Improvement

Sample Output

Employees Scoring 80 or Above:

E101 Anuj 92

E104 Neha 88

Employees Needing Improvement: 2

Highest Performer:

E101 Anuj 92

High Performers:

['Anuj', 'Rahul', 'Neha']

Performance Categories:

Anuj -> Excellent

Rahul -> Good

Priya -> Needs Improvement

Neha -> Good

Amit -> Needs Improvement

Question 2: Flight Booking Analysis

Problem Statement

A flight reservation system stores passenger records as tuples:

```
bookings = (  
    ("P101", "Delhi", "Confirmed"),  
    ("P102", "Mumbai", "Waiting"),  
    ("P103", "Delhi", "Confirmed"),  
    ("P104", "Chennai", "Cancelled"),  
    ("P105", "Mumbai", "Confirmed"),  
    ("P106", "Delhi", "Waiting")  
)
```

Where:

- Passenger ID
- Destination
- Booking Status

Tasks

Write a Python program to:

1. Display all passengers whose booking status is **Confirmed**.
2. Count the number of passengers travelling to **Delhi**.
3. Count Confirmed, Waiting, and Cancelled bookings separately.
4. Create a list containing passenger IDs with Waiting status.
5. Determine which destination has the highest number of bookings.

Sample Output

Confirmed Passengers:

P101 Delhi

P103 Delhi

P105 Mumbai

Passengers Travelling to Delhi: 3

Confirmed: 3

Waiting: 2

Cancelled: 1

Waiting List:

['P102', 'P106']

Most Booked Destination:

Delhi